SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	**Assessment:**	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	**Total Marks:**	100
CO5 Statement:	*Construct MST using shortest path algorithms and traverse the graph to model and analyse realworld problem.*		
Student Name:	RAGUL.M	**Reg. No.:**	192524066
Section / Batch:	SLOT A /CSA0305	**Date:**	31/08/2026

Code of Conduct

I _RAGUL.M/192524066 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: RAGUL.M

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20

Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
	GRAND TOTAL:	**100 Marks**	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or nonfunctional code
Code Quality & Readability	20	Clean, modular, wellcommented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	Course Coordinator	Program Director
Signature: ____________	Signature: ____________	Signature: ____________
Date: ________________	Date: ________________	Date: ________________

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QUESTIONS

Q1
Minimum Spanning Tree - Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q1 Minimum Spanning Tree - Kruskal's Algorithm 20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

a. All zones must be connected

b. Total installation cost must be minimized

c. No redundant connections (no cycles) allowed

d. The network must remain fully connected

Your Answer

```
STEP 2: Sort all edges
   Sort all edges E in non-decreasing order of installation cost (weight)

STEP 3: Process edges one by one
   edgeCount = 0
   FOR each edge (u, v, cost) in sorted E:
```

Save Answer Next →

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20 marks

Q2
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20 marks

Q3
Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q2 Graph - Topological Sort 20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.

Constraints:

a. Detect cycle before sorting

b. Use DFS approach

Your Answer

```
   Output "Cycle detected: Topological Sort not possible"
Else:
   Pop all elements from Stack and output them
```

Save Answer Next →

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QUESTIONS

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Q3
Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q3 **Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's** 20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.
Constraints:
a. Use MST for infrastructure
b. Use shortest path for routing

Your Answer

```
Step 5: Compute shortest paths from a given source location
INPUT Source Location S
ShortestPaths = DijkstraShortestPath(Infrastructure, S)
PRINT "Shortest Routing Paths from Source:", S
FOR each vertex v in Infrastructure:
   PRINT "Distance to", v, "=", ShortestPaths[v]
```

Save Answer | Next →

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QUESTIONS

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20 marks

Q2
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20 marks

Q3
Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q4 **Shortest Path Algorithm - Dijkstra's Algorithm** 20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

```
        if freeDist < dist[v][1]:
            dist[v][1] = freeDist
            PQ.insert((freeDist, v, 1))
answer = MIN(dist[D][0], dist[D][1])
return answer
END
```

Save Answer | Next →

QUESTIONS

Q1
Minimum Spanning Tree - Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm

20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST. Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

```
SECOND_BEST_MST(G):
   T = MST(G)
   Wbest = total_weight(T)
   Wsecond = infinity

for each edge e in T:
```

Save Answer